CURRICULUM VITAE · ACADEMIC

Yi Xia *(Sean)*

PhD Candidate in Computer Vision & Human Activity Recognition · Auckland University of Technology

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8	2	623 h
FIRST-AUTHOR PAPERS	Q1 ARTICLES	TEACHING (TA)

RESEARCH PROFILE

I am a PhD candidate at **Auckland University of Technology (AUT)** working on deep learning, computer vision and graph neural networks. My thesis is submitted and the oral examination is in February 2027. I am first author on all eight of my peer-reviewed publications, including two Q1 journal articles on skeleton-based human activity recognition: a frequency-aware model in *Pattern Recognition* (IF 7.6) and a generalisation-centred review in *Computer Science Review* (IF 17.5). A further paper, ESA-GCN, is accepted at ICONIP 2026. My most-cited paper, on kiwifruit detection, has **50+ citations**. Alongside this research I built production computer-vision systems at TOHU MEDIA and an offline LLM digital-forensics application for Tonga Police. Next, I plan to work where computer vision and large language models meet: using language and vision-language models to make activity recognition hold up beyond benchmarks, and building privacy-aware LLM systems that run on local hardware.

RESEARCH INTERESTS

- Skeleton-based human activity recognition
- Real-time object detection & tracking
- Frequency-domain representation learning
- Private LLM systems & digital-forensics analysis
- Spatio-temporal graph convolutional networks
- Video analytics for agriculture & media
- Attention & Transformer architectures for vision
- Practical, privacy-aware AI deployment

EDUCATION

Doctor of Philosophy (PhD), Computer & Information Sciences — *Auckland University of Technology* — Oct 2023 – present

Thesis: *Deep Learning for Skeleton-Based Human Activity Recognition*. Submitted; oral examination February 2027.
Supervisors: **Dr Raymond Lutui** and **Associate Professor Sira Yongchareon**. **AUT PhD Full Scholarship**.

Master of Computer & Information Sciences (Computer Vision) — *Auckland University of Technology* — 2021 – 2023

Research thesis: *Kiwifruit Detection and Tracking from a Deep-Learning Perspective Using Digital Videos*.

Bachelor of Engineering, Logistics Engineering (Software Development) — *Dalian Neusoft University of Information, China* — 2011 – 2015

TEACHING EXPERIENCE

Teaching Assistant — *School of Engineering, Computer & Mathematical Sciences, AUT* — Mar 2025 – present

- Deliver classes and tutorials, prepare teaching materials and mark assessment across seven courses (NZQF Levels 5–8), with 623 formally allocated hours over four consecutive semesters.

Co-Founder — *Baiyin Pinuo Training School, China* — 2019 – 2021

- Co-founded a supplementary-education school; designed curriculum and led academic operations.

Junior Lecturer, Department of Information Engineering — *Baiyin Vocational & Technology College, China* — Aug 2015 – May 2019

- Designed and taught computing, Java, IoT and RFID courses (lectures and labs) to about 120 students a year.

PUBLICATIONS

Reverse chronological. First author on all eight peer-reviewed publications. Live citation counts on Google Scholar.

JOURNAL ARTICLES

01 **Xia, Y.**, Yongchareon, S., Lutui, R., & Sheng, Q. Z. (2026). Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition. *Computer Science Review*, 62, 101030. Q1 · IF = 17.5
https://doi.org/10.1016/j.cosrev.2026.101030

02 **Xia, Y.**, Yongchareon, S., Lutui, R., & Sheng, Q. Z. (2026). Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition. *Pattern Recognition*, 175, 113146. Q1 · IF = 7.6
https://doi.org/10.1016/j.patcog.2026.113146

BOOK CHAPTERS

01 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2025). An Improved YOLO Model for Kiwifruit Detection. In *Optimization, Machine Learning, and Fuzzy Logic* (Advances in Computational Intelligence and Robotics series), IGI Global.
https://doi.org/10.4018/979-8-3693-7352-1.ch008

REFEREED CONFERENCE PAPERS

01 **Xia, Y.**, Yongchareon, S., & Lutui, R. (2026). ESA-GCN: Efficient Semantic-Aware Graph Convolution for Skeleton-Based Action Recognition. In *Proceedings of the International Conference on Neural Information Processing (ICONIP 2026)*, Springer LNCS. Accepted, to appear.
https://shawn7702.github.io/esa-gcn/

02 **Xia, Y.**, Yongchareon, S., Lutui, R., & Sheng, Q. Z. (2025). Dynamic Self-Attention Gated Spatial-Temporal Graph Convolutional Network for Skeleton-Based Human Activity Recognition. In *Proceedings of the 31st International Conference on Neural Information Processing (ICONIP 2024)*, Springer LNCS.
https://doi.org/10.1007/978-981-96-6588-4_14

03 **Xia, Y.**, Nguyen, M., Lutui, R., & Yan, W. Q. (2023). Multiscale Kiwifruit Detection from Digital Images. In *Proceedings of the Pacific-Rim Symposium on Image and Video Technology (PSIVT 2023)*, Springer.
https://doi.org/10.1007/978-981-97-0376-0_7

04 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2023). Kiwifruit Counting Using KiwiDetector and KiwiTracker. In *Proceedings of the SAI Intelligent Systems Conference 2023*, Springer Lecture Notes in Networks and Systems.
https://doi.org/10.1007/978-3-031-47724-9_41

05 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2022). A Real-Time Kiwifruit Detection Based on Improved YOLOv7. In *Proceedings of the 37th International Conference on Image and Vision Computing New Zealand (IVCNZ 2022)*, Springer LNCS. 50+ cites
https://doi.org/10.1007/978-3-031-25825-1_4

THESES

01 **Xia, Y.** (2023). *Kiwifruit Detection and Tracking from a Deep-Learning Perspective Using Digital Videos. Master's thesis*, Auckland University of Technology.

CONFERENCE PRESENTATIONS

- "ESA-GCN: Efficient Semantic-Aware Graph Convolution for Skeleton-Based Action Recognition," poster presentation, *ICONIP 2026*, Melbourne, 23 November 2026 (scheduled).
- "Dynamic Self-Attention Gated Spatial-Temporal GCN for HAR," oral presentation, *ICONIP 2024*.
- "Multiscale Kiwifruit Detection from Digital Images," oral presentation, *PSIVT 2023*, Auckland, November 2023.
- "Kiwifruit Counting Using KiwiDetector and KiwiTracker," presentation, *SAI Intelligent Systems Conference*, 2023.
- "A Real-Time Kiwifruit Detection Based on Improved YOLOv7," oral presentation, *IVCNZ 2022*, Auckland, October 2022.

RESEARCH & APPLIED AI EXPERIENCE

Doctoral Researcher — *Institute of Robotics & Vision (IoRV), AUT* Oct 2023 – present

- Designed graph-convolutional models for skeleton-based HAR: FATL-GCN (*Pattern Recognition*, 2026), a dynamic self-attention gated ST-GCN (ICONIP 2024) and ESA-GCN (ICONIP 2026); trained and evaluated on NTU RGB+D 60/120 and related benchmarks.

Research Assistant, LLM Digital Forensics — *Auckland University of Technology* Oct 2025 – Feb 2026

Client: Tonga Police.

- Built and delivered Autopsy AI, a Windows application that parses Autopsy HTML reports, combines rule-based risk scoring with LLM analysis, and generates investigator-ready reports.
- Designed a fully offline Ollama/Gemma deployment with English/Tongan risk keywords, multi-provider LLM support and a standalone executable to keep forensic data on-device.

Research Assistant — *School of Engineering, Computer & Mathematical Sciences, AUT* Nov 2022 – Jul 2023 (multiple appointments)

- Funded by the AUT Summer RA Scholarship and SECMS Research Assistant Fund.
- Co-developed KiwiDetector and KiwiTracker for orchard video; contributed to three peer-reviewed publications.

INDUSTRY EXPERIENCE

Artificial Intelligence Developer — *TOHU MEDIA · Tāmaki Makaurau / Auckland* Feb 2022 – May 2026

Māori-owned media-production company. Full-time from June 2025 to May 2026; previously part-time alongside master's and PhD studies.

- Owned the company's AI capability end-to-end; shipped **three production systems for object detection, OCR and action recognition** on client video footage (specific deliverables under NDA).
- Engineered for variable cinematography, dense overlay text, low-contrast archives, long-form takes and workloads beyond a single-workstation GPU.
- Owned dataset scope, PyTorch training/evaluation, **Docker deployment to cloud GPUs**, end-user iteration and data sovereignty on company-controlled infrastructure.

AWARDS & SCHOLARSHIPS

- **AUT PhD Full Scholarship** — tuition fees and living stipend (Oct 2023 – Oct 2026).
- **SECMS Research Assistant Fund** — May 2023, AUT.
- **AUT Summer Research Assistant Scholarship** — December 2022.

PROFESSIONAL SERVICE & MEMBERSHIPS

- Reviewer, *ACM Transactions on Multimedia Computing, Communications, and Applications* (TOMM), 2024.
- Reviewer, *International Agrophysics* (2025, 2026) and IEEE PIMRC (2026).
- Member, AUT Institute of Robotics & Vision (IoRV), 2022 – present.

TECHNICAL SKILLS

Programming	Python (8+ years) · Java (taught for 4 years).
Deep-learning frameworks	PyTorch (primary) · TensorFlow / Keras · Hugging Face.
CV & scientific computing	OpenCV · scikit-learn · NumPy · pandas · SciPy · Matplotlib.
Methods	CNNs · GCNs · Transformers / attention · object detection · tracking · action recognition · OCR · segmentation · skeleton-based HAR · frequency-domain analysis.
LLM systems	Ollama/Gemma · Claude/OpenAI APIs · offline inference · multi-provider integration · prompt engineering.
Deployment & tooling	Docker · Windows/Linux · cloud GPU inference · standalone apps · Git/GitHub · CUDA · Jupyter.

LANGUAGES

Mandarin Chinese	**English**
Native	*Professional working proficiency*

REFEREES

- **Dr Raymond Lutui**, AUT, primary doctoral supervisor · raymond.lutui@aut.ac.nz
- **Associate Professor Sira Yongchareon**, AUT, doctoral supervisor · sira.yongchareon@aut.ac.nz